

Module 4: Linear regression/Generalized linear regression

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Outline

In this module, we will review linear regression.

Linear regression

- Model:

$$Y_{n \times 1} = X_{n \times p} \beta_{p \times 1} + \epsilon_{n \times 1}$$

- Equivalently:

$$y_i = x_i^T \beta + \epsilon_i, \quad i = 1, \dots, n$$

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- Standard assumptions

- y_i independent (equivalently ϵ_i independent)
- $\mathbb{E}(\epsilon_i) = 0$
- $\text{var}(\epsilon_i) = \sigma^2$, constant
- x_i known, β to be estimated

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- More concisely:

$$\mathbb{E}(Y \mid X) = X\beta, \quad \text{var}(Y \mid X) = \sigma^2 I$$

Interpretation of β_j

- Effect on the expected response of a unit change in j th explanatory variable, all other variables held fixed

Least squares estimation

- Definition (minimize the residuals)

$$\hat{\beta}_{\text{LS}} := \min_{\beta} \sum_{i=1}^n (y_i - x_i^T \beta)^2$$

- Equivalently,

$$\hat{\beta}_{\text{LS}} := \min_{\beta} (y - X\beta)^T (y - X\beta)$$

- Equivalently (L2 distance),

$$\hat{\beta}_{\text{LS}} := \min_{\beta} \|y - X\beta\|_2^2$$

- Equivalently, $\hat{\beta}$ is the solution of the score equation

$$X^T(y - X\beta) = 0$$

- Solution

$$\hat{\beta}_{\text{LS}} = (X^T X)^{-1} (X^T y)$$

Another interpretation: the projection of $\mathbf{Y}$ onto the linear subspace spanned by the columns of $\mathbf{X}$

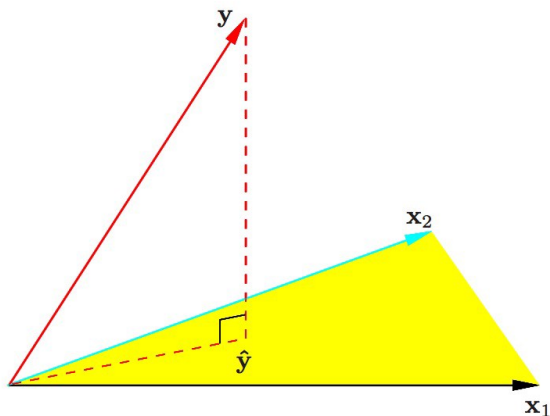


FIGURE 3.2. The N -dimensional geometry of least squares regression with two predictors. The outcome vector $\mathbf{y}$ is orthogonally projected onto the hyperplane spanned by the input vectors $\mathbf{x}_1$ and $\mathbf{x}_2$. The projection $\hat{\mathbf{y}}$ represents the vector of the least squares predictions

Least squares estimation (cont'd)

Assume X is fixed,

- Expected value

$$\mathbb{E}(\hat{\beta}_{LS}) = (X^T X)^{-1} X^T \mathbb{E}(y) = (X^T X)^{-1} (X^T X) \beta = \beta$$

- Variance

$$\begin{aligned}\text{var}(\hat{\beta}_{LS}) &= (X^T X)^{-1} X^T \text{var}(y) X (X^T X)^{-1} \\ &= (X^T X)^{-1} X^T \sigma^2 I X (X^T X)^{-1} \\ &= \sigma^2 (X^T X)^{-1}\end{aligned}$$

Assumptions for ordinary least squares

- **Linearity**: the expectation of Y is linear in $X_1 \dots X_p$
- **Independence**: the ϵ_i are independent
- **Mean zero errors**: the ϵ_i have mean zero, i.e. $E[\epsilon_i] = 0$
- **Equal variance (homoscedasticity)**: the ϵ_i have the same variance, i.e. $\text{Var}[\epsilon_i] = \sigma^2$

What about normal distribution?

- If we further assume $\epsilon_i \sim N(0, \sigma^2)$ (and independent across i), then
- $y \mid X \sim N(X\beta, \sigma^2 I)$, and
- likelihood function is

$$L(\beta, \sigma^2; y) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp \left\{ -\frac{1}{2\sigma^2} (y - X\beta)^T (y - X\beta) \right\}$$

- log-likelihood function is

$$\ell(\beta, \sigma^2; y) = -\frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} (y - X\beta)^T (y - X\beta)$$

- maximum likelihood estimate of β is

$$\hat{\beta}_{ML} = (X^T X)^{-1} X^T y = \hat{\beta}_{LS}$$

What about normal distribution? (cont'd)

- distribution of $\hat{\beta}$ is normal

$$\hat{\beta} \sim N_p \left(\beta, \sigma^2 (X^T X)^{-1} \right)$$

- distribution of $\hat{\beta}_j$ is

$$N \left(\beta_j, \sigma^2 (X^T X)^{-1}_{jj} \right), \quad j = 1, \dots, p$$

- maximum likelihood estimate of σ^2 is

$$\frac{1}{n} (y - X\hat{\beta})^T (y - X\hat{\beta})$$

- but we use

$$\tilde{\sigma}^2 = \frac{1}{n - p} (y - X\hat{\beta})^T (y - X\hat{\beta})$$

Maximum likelihood estimation vs. OLS

- We did not place any distributional assumptions on the outcome,
 - We only required that $E[\epsilon_i] = 0$ with constant variance
 - In other words, OLS is a semiparametric method

Maximum likelihood estimation vs. OLS

- We did not place any distributional assumptions on the outcome,
 - We only required that $E[\epsilon_i] = 0$ with constant variance
 - In other words, OLS is a semiparametric method
- Sometimes, people assume that $\epsilon_i \sim N(0, \sigma^2)$, which means

$$Y_i \sim N(\beta_0 + \beta_1 X_{i1} + \dots + \beta_p X_{ip}, \sigma^2)$$

- If this additional assumption is made, then we can instead use maximum likelihood estimation for β
- This connects to a whole other class of models called generalized linear models (GLMs)
- Interestingly, in this case, you will end up with the same estimates for β

Resources

This tutorial is based on

- Nancy Reid's STA2101 Methods of Applied Statistics [links]
- Harvard's Biostatistics Preparatory Course Methods [links].

Break



Outline

In this module, we will review generalized linear regression.

Logistic regression

- Each response is binary: $y_i = 1, 0$
- Explanatory variables x_i^T as usual
- Model

$$P(y_i = 1 \mid x_i) = \frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)}$$

Generalized linear models (GLMs)

- Generalized Linear Models extend the classical set-up to allow for a wider range of distributions
- GLMs have three pieces
 - ① random component: $y_i \sim$ some distribution with $E[y_i | \mathbf{x}_i] = \mu_i$
 - ② systematic component: $\mathbf{x}_i^T \beta$
 - ③ The link function that links the random and systematic components
 $g(u_i) = \mathbf{x}_i^T \beta$
- Distributions of y_i comes from exponential family.

Exponential family

The random variable Y belongs to the exponential family of distributions if its support does not depend upon any unknown parameters and its density or probability mass function takes the form

$$f(y \mid \theta, \phi) = \exp \left(\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right)$$

where θ is the canonical parameter (related to the mean), ϕ is a dispersion parameter (often related to the variance), $b(\theta)$ is the cumulant function (which helps derive the mean and variance), and $c(y, \phi)$ is a normalization term ensuring the density integrates (or sums) to 1.

Example 1 Gaussian distribution

The Gaussian (Normal) distribution can be written in exponential family form as:

$$f(y \mid \mu, \sigma^2) = \exp \left(\frac{y\mu - \frac{\mu^2}{2}}{\sigma^2} - \frac{y^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2) \right)$$

where:

$$\theta = \mu \quad \text{(natural parameter)}$$

$$\phi = \sigma^2 \quad \text{(dispersion parameter)}$$

$$b(\theta) = \frac{\theta^2}{2} \quad \text{(cumulant function)}$$

$$c(y, \phi) = -\frac{y^2}{2\phi} - \frac{1}{2} \log(2\pi\phi) \quad \text{(normalization term)}$$

Example 2 Poisson distribution

The Poisson distribution with parameter λ (rate parameter) can be written in exponential family form as:

$$f(y \mid \lambda) = \exp(y \log \lambda - \lambda - \log(y!))$$

where:

$$\theta = \log \lambda \quad (\text{natural parameter})$$

$$\phi = 1 \quad (\text{dispersion parameter})$$

$$b(\theta) = e^{\theta} \quad (\text{cumulant function})$$

$$c(y, \phi) = -\log(y!) \quad (\text{normalization term})$$

Example 3 Binomial distribution

The Binomial distribution with parameters n (number of trials) and p (success probability) can be written in exponential family form as:

$$f(y | p) = \exp \left(y \log \left(\frac{p}{1-p} \right) + n \log(1-p) + \log \binom{n}{y} \right)$$

where:

$$\theta = \log \left(\frac{p}{1-p} \right) \quad (\text{natural parameter, log-odds})$$

$$\phi = 1 \quad (\text{dispersion parameter})$$

$$b(\theta) = n \log(1 + e^\theta) \quad (\text{cumulant function})$$

$$c(y, \phi) = \log \binom{n}{y} \quad (\text{normalization term})$$

MGF of exponential family

$$\begin{aligned}M_Y(t) &= \mathbb{E}[e^{tY}] \\&= \int e^{ty} f(y \mid \theta, \phi) dy \\&= \int \exp \left(ty + \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right) dy \\&= \int \exp \left(\frac{y(\theta + ta(\phi)) - b(\theta)}{a(\phi)} + c(y, \phi) \right) dy \\&= \exp \left(\frac{b(\theta + ta(\phi)) - b(\theta)}{a(\phi)} \right) \int \exp \left(\frac{y(\theta + ta(\phi)) - b(\theta + ta(\phi))}{a(\phi)} + c(y, \phi) \right) dy \\&= \exp \left(\frac{b(\theta + ta(\phi)) - b(\theta)}{a(\phi)} \right)\end{aligned}$$

Mean of the exponential family

$$\begin{aligned}M'_Y(t) &= \frac{d}{dt} \left[\exp \left(\frac{b(\theta + ta(\phi)) - b(\theta)}{a(\phi)} \right) \right] \\&= M_Y(t) \cdot \frac{d}{dt} \left(\frac{b(\theta + ta(\phi)) - b(\theta)}{a(\phi)} \right) \quad (\text{Chain rule}) \\&= M_Y(t) \cdot \frac{b'(\theta + ta(\phi)) \cdot a(\phi)}{a(\phi)} \\&= M_Y(t) \cdot b'(\theta + ta(\phi))\end{aligned}$$

Evaluating at $t = 0$ (since $M_Y(0) = 1$):

$$M'_Y(0) = b'(\theta) = \mu = \mathbb{E}[Y]$$

Variance of the exponential family

$$\begin{aligned}M_Y''(t) &= \frac{d}{dt} [M_Y(t) \cdot b'(\theta + ta(\phi))] \\&= M_Y'(t) \cdot b'(\theta + ta(\phi)) + M_Y(t) \cdot b''(\theta + ta(\phi)) \cdot a(\phi) \\&= M_Y(t) \cdot \left[(b'(\theta + ta(\phi)))^2 + b''(\theta + ta(\phi)) \cdot a(\phi) \right]\end{aligned}$$

Evaluating at $t = 0$:

$$M_Y''(0) = \mu^2 + b''(\theta)a(\phi) = \mathbb{E}[Y^2]$$

Thus the variance is:

$$\text{Var}(Y) = M_Y''(0) - [M_Y'(0)]^2 = b''(\theta)a(\phi)$$

Link function

The second element of the generalization is that instead of modeling the mean, as before, we will introduce a one-to-one continuous differentiable transformation $g(\mu_i)$ of the mean $\mu_i = E[y_i]$ and model that

$$g(\mu_i) = \eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$$

where η_i is the linear predictor. The function $g(\cdot)$ is called the link function.

Link function

Since $g(\cdot)$ is a one-to-one transformation, we can invert it to get the mean:

$$\mu_i = g^{-1}(\eta_i) = g^{-1}(\mathbf{x}_i^\top \boldsymbol{\beta})$$

Note that we do not transform the response variable y_i itself, but rather the mean of the response variable.

Link function in R

“glm” has several options for family:

```
binomial(link = "logit")  
gaussian(link = "identity")  
Gamma(link = "inverse")  
inverse.gaussian(link = "1/mu^2")  
poisson(link = "log")  
quasi(link = "identity", variance = "constant")  
quasibinomial(link = "logit")  
quasipoisson(link = "log")
```

An important practical feature of generalized linear models is that they can all be fit to data using the same algorithm, a form of iteratively re-weighted least squares (IRLS).

- 1 Initialize $\hat{\mu}_i^{(0)} = y_i + \epsilon$, $\eta_i^{(0)} = g(\hat{\mu}_i^{(0)})$
- 2 While not converged:
 - Working response: $z_i = \eta_i^{(k)} + (y_i - \hat{\mu}_i^{(k)}) \left(\frac{d\eta}{d\mu} \right)$
 - Weights: $w_i = \left[V(\hat{\mu}_i^{(k)}) \left(\frac{d\mu}{d\eta} \right)^2 \right]^{-1}$
 - Update: $\beta^{(k+1)} = (X^\top W X)^{-1} X^\top W z$
 - Update $\eta_i^{(k+1)}$, $\hat{\mu}_i^{(k+1)}$

Gaussian Special Case of IRLS

For linear regression ($Y \sim N(\mu, \sigma^2)$):

- **Link:** Identity $g(\mu) = \mu$
- **Variance:** $V(\mu) = 1$ (constant)
- **Weights:** $w_i = 1$ (equal weighting)
- **Working response:** $z_i = y_i$ (original data)

IRLS reduces to ordinary least squares:

$$\beta^{(k+1)} = (X^T X)^{-1} X^T y \quad (\text{single iteration})$$

Key observations:

- Link derivative $\frac{d\mu}{d\eta} = 1$
- No reweighting needed (homoscedasticity)
- Exact solution in one step

Asymptotics

The asymptotic distribution of the MLE $\hat{\beta}$ is given by:

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, (X^\top W X)^{-1} \phi)$$

In the case of the Gaussian distribution, ϕ is the variance σ^2 , W is the identity matrix, and the covariance matrix simplifies to: $(X^\top X)^{-1} \sigma^2$.